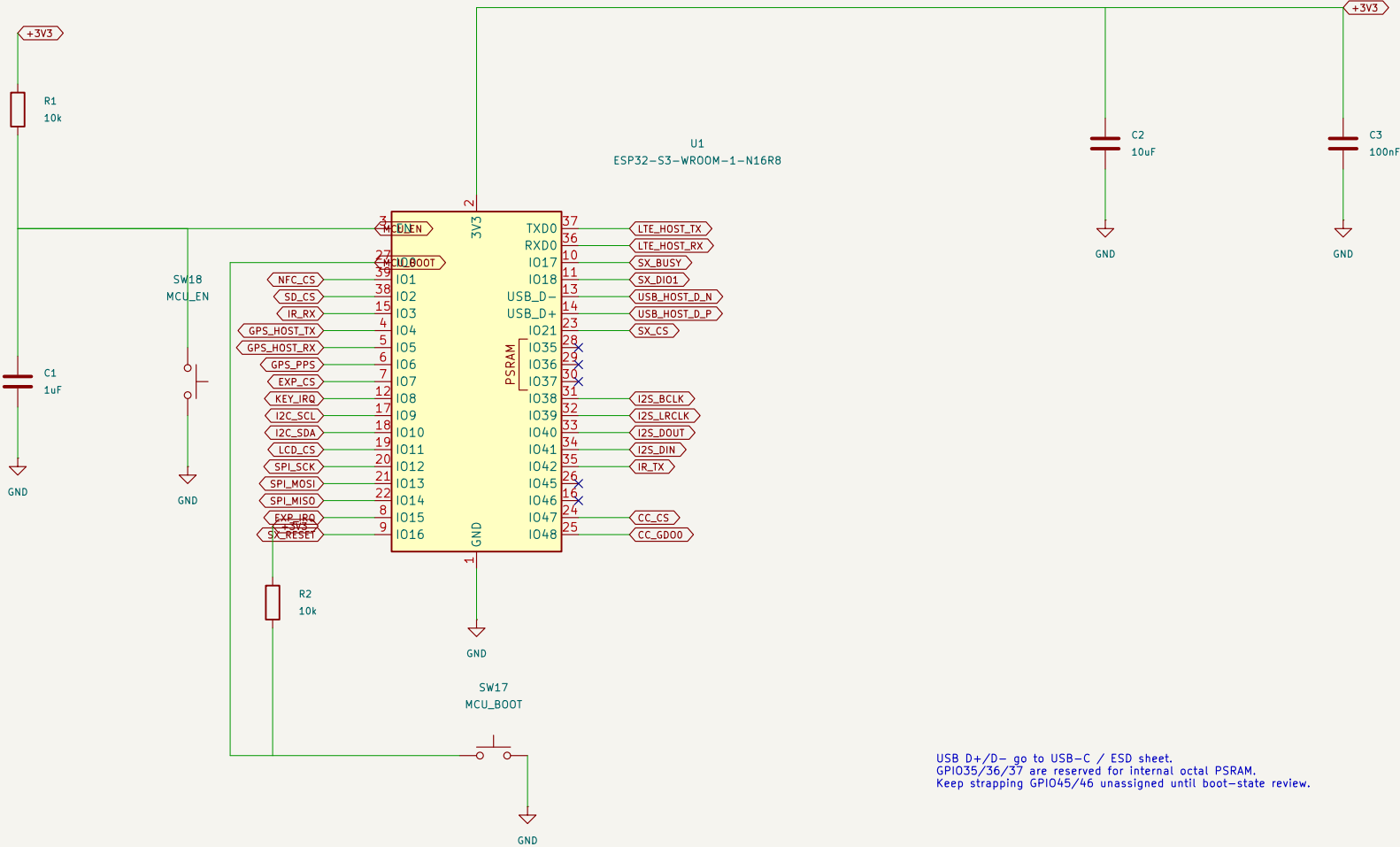


	1	2	3	4	5	6	7	8	9	10	11	
	HANDSET REV A / SYSTEM SCHEMATIC INDEX											
	Engineering draft. Retained ESP32-S3 / MakerFocus 3000 mAh / external display / all peripheral interfaces.											
A	ESP32 host, boot and reset				Key matrix and trackball				USB-C data and ESD			
	File: core.kicad_sch				File: keys.kicad_sch				File: usb.kicad_sch			
B	Fused USB Input and efuse				USB-C source detection and permission				Battery, thermistor and programmable charger			
	File: usb-input.kicad_sch				File: usb-detect.kicad_sch				File: power.kicad_sch			
C	Power control and fuel gauge				Main 3.3 V buck-boost supply				Auxiliary 5 V and 1.8 V supplies			
	File: power-control.kicad_sch				File: reg3.kicad_sch				File: aux-power.kicad_sch			
D	Regulated modem supply and enable				SX1262 LoRa radio				CC1101 sub-GHz radio			
	File: modem-power.kicad_sch				File: radios.kicad_sch				File: cc-radio.kicad_sch			
E	125 kHz reader analog circuit				LF reader logic translation				GNSS receiver and passive antenna			
	File: lf-radio.kicad_sch				File: lf-support.kicad_sch				File: gnss-antennas.kicad_sch			
F	NFC reader, oscillator and local supplies				Media speaker amplifier and digital microphone				Cellular adapter and call-audio harnesses			
	File: nfc.kicad_sch				File: audio.kicad_sch				File: interfaces.kicad_sch			
G	Protected external module port				MicroSD and case-mounted infrared				External display, touch and backlight			
	File: protection.kicad_sch				File: storage-ir.kicad_sch				File: display.kicad_sch			
H	Always-on charging supervisor				Display supply and translation				Expansion control and isolation			
	File: power-supervisor.kicad_sch				File: display-interface.kicad_sch				File: expansion-control.kicad_sch			
	Sheet: / File: handset.kicad_sch											
	Title:											
	Size: A2										Date:	Rev:
	KiCad E.D.A. 10.0.6										Id: 1/25	
	1	2	3	4	5	6	7	8	9	10	11	

ESP32 host, boot and reset

Retain N16R8. Module pads 28/29/30 belong to octal PSRAM. USB programming uses the USB sheet.



USB D+/D- go to USB-C / ESD sheet.
GPIO35/36/37 are reserved for internal octal PSRAM.
Keep strapping GPIO45/46 unassigned until boot-state review.

RESET / EN: 10k pull-up + 1uF
BOOT held low during reset selects download mode.

Circuit review in progress – not released for fabrication

Sheet: /ESP32 host, boot and reset/
File: core.kicad_sch

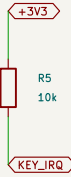
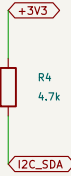
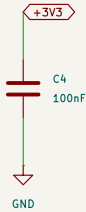
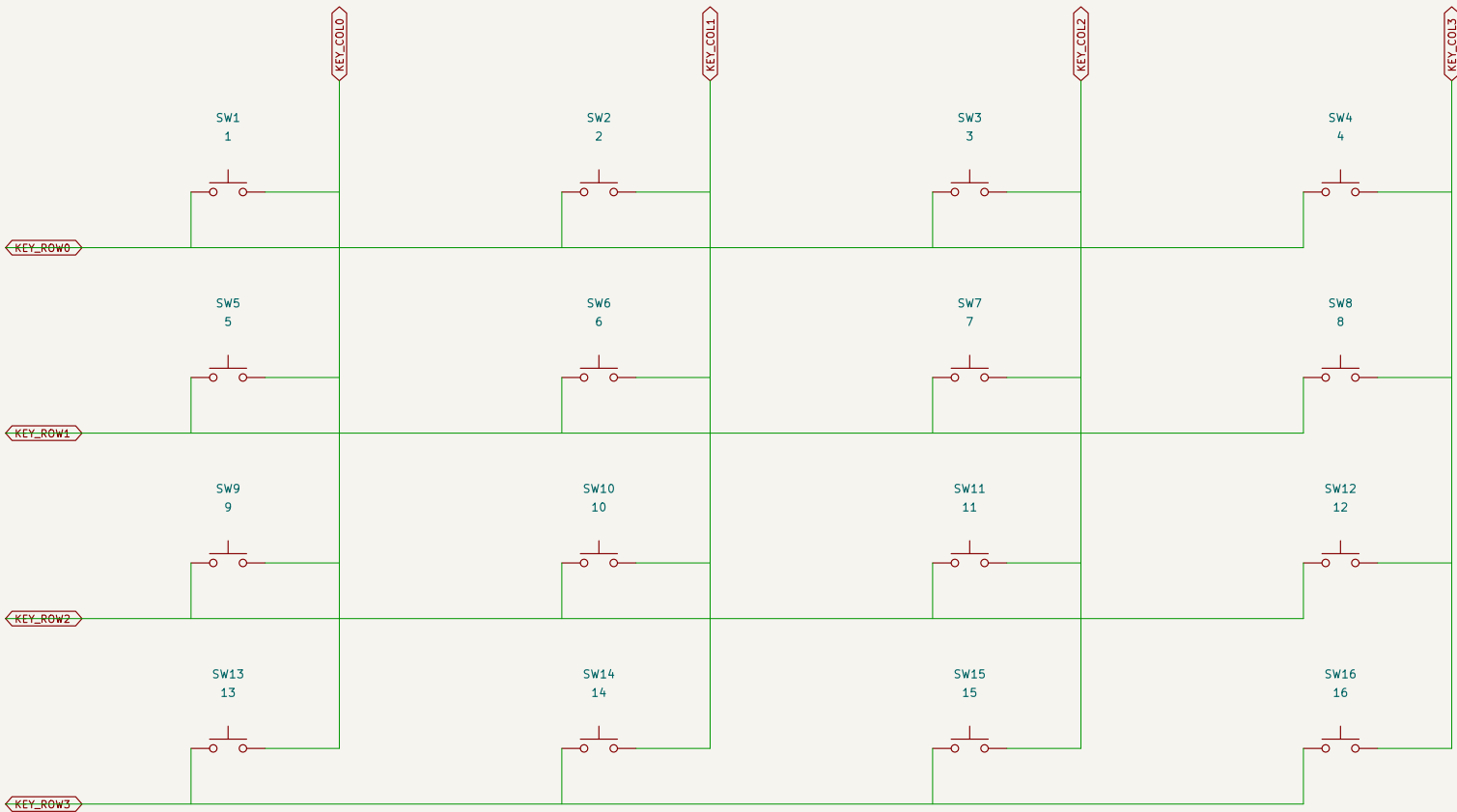
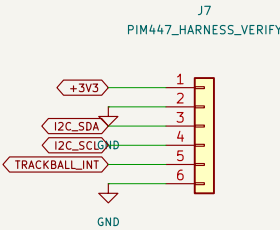
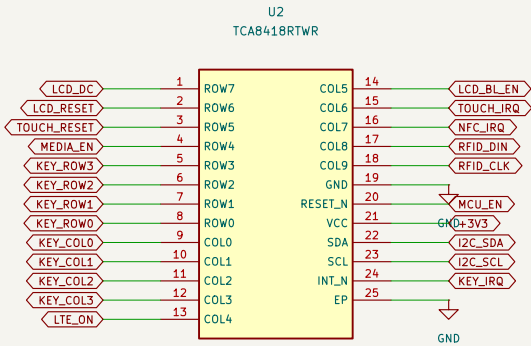
Title: ESP32 host, boot and reset

Size: A3
Date: 2026-09-24

Rev: A – engineering draft
Id: 2/25

Key matrix and trackball

TCAB418: 4 rows x 4 columns; remaining pins are firmware-configured GPIO. Confirm PIM447 harness orientation.



Circuit review in progress – not released for fabrication

Sheet: /Key matrix and trackball/
File: keys.kicad_sch

Title: Key matrix and trackball

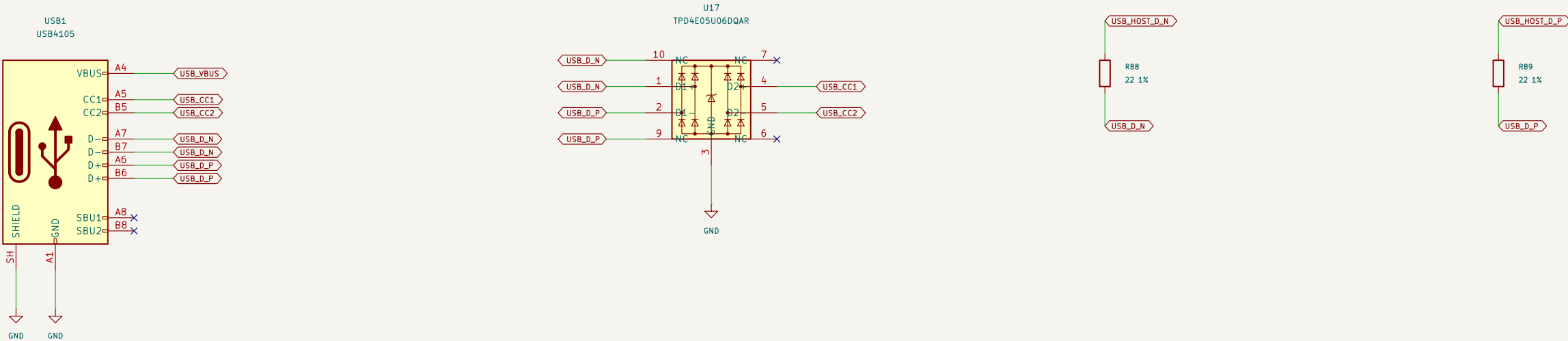
Size: A3
Date: 2026-09-24

KiCad E.D.A. 10.0.6

Rev: A – engineering draft
Id: 3/25

USB-C data and ESD

U17 NC lands 10/9 externally bridge to 1/2 for pass-through routing; no internal connection. R88/R89 host series termination, initial 22 ohms; qualify on hardware.



Circuit review in progress – not released for fabrication

Sheet: /USB-C data and ESD/
File: usb.kicad_sch

Title: USB-C data and ESD

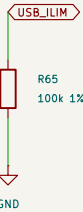
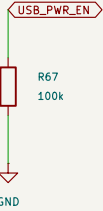
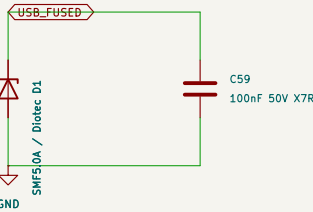
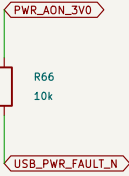
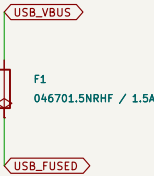
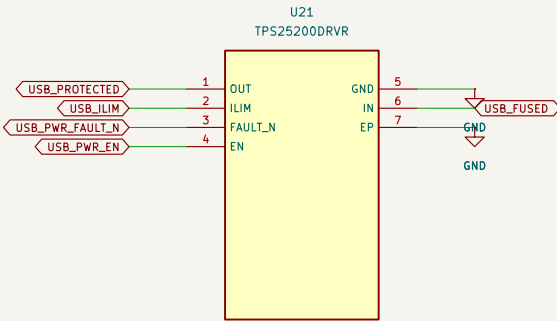
Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 4/25

Fused USB input and eFuse

TPS25200 R_ILIM = 100k, 1%. F1/TVS pulse coordination requires bench validation. No USB PD negotiation.



Supply flag: external USB VBUS through F1. No internal source is implied.

Circuit review in progress – not released for fabrication

Sheet: /Fused USB input and eFuse/
File: usb-input.kicad_sch

Title: Fused USB input and eFuse

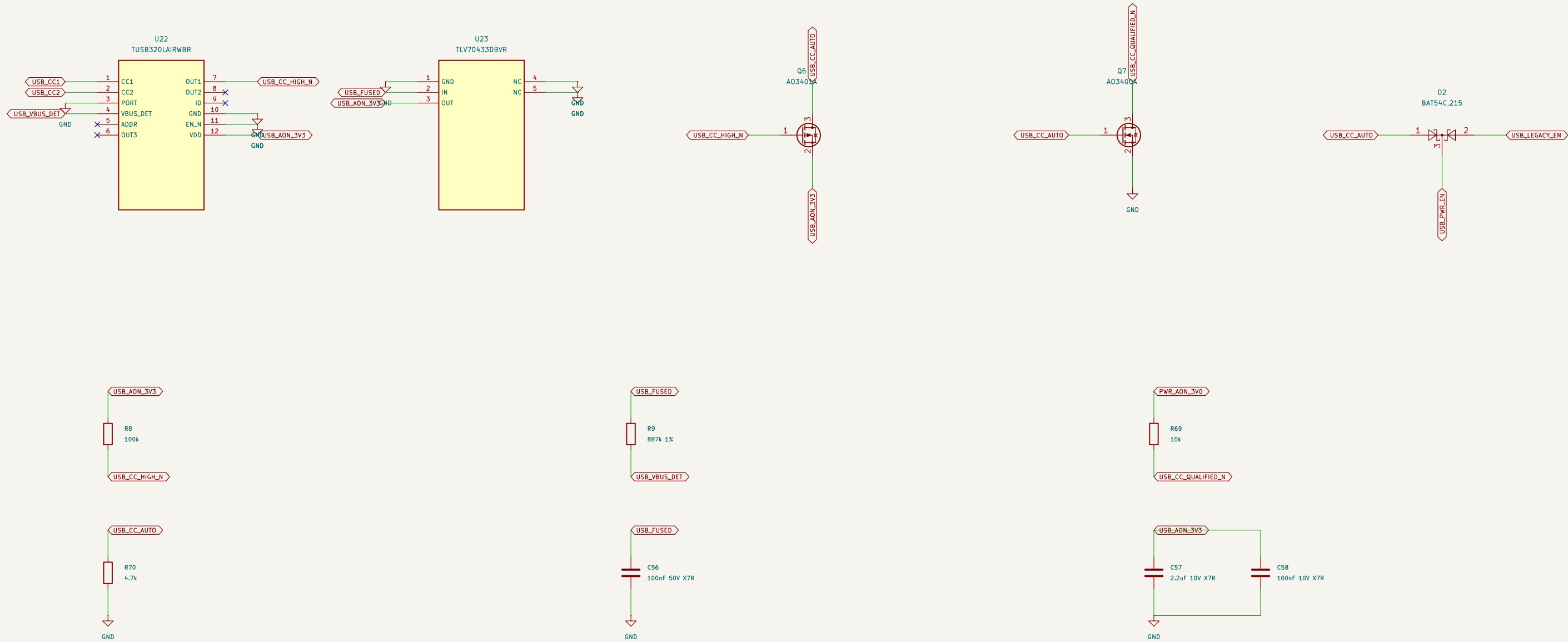
Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 5/25

USB-C source detection and permission

Sink-only internal Rd. OUT1 low authorizes Type-C 1.5A/3A. Legacy permission requires USB enumeration; see charger-policy.json.



Battery, thermistor and programmable charger

Verify actual battery polarity. J2 is a battery-mounted 10k B3435 NTC. CE defaults OFF. Firmware must apply charger-policy.json.

Charge enable: default OFF
CHG_ENABLE alone cannot charge; U25 CHG_ARM is also required.

Battery-mounted thermistor
Do not substitute a fixed resistor.

Protected 103665 pack / 3000 mAh
F2 does not enforce the pack operating-current limit.

Q5 AND Q8 must both conduct to enable charging.
U25 PA0 reset/high-Z -> R79 pulls Q8 OFF.
Preload PA0 low; arm only after verified charger policy.
IWDG reset and GPIO discharge timing require bench qualification.
Supply flags identify the external protected pack through F2 and its return.

Circuit review in progress – not released for fabrication

Sheet: /Battery, thermistor and programmable charger/
File: power.kicad_sch

Title: Battery, thermistor and programmable charger

Size: A3	Date: 2026-09-24	Rev: A – engineering
KiCad E.D.A. 10.0.6		Id: 7/25

Engineering draft / named nets are common / NC = intentional unused pin

Battery, thermistor and programmable charger

Verify actual battery polarity. J2 is a battery-mounted 10k B3435 NTC. CE defaults OFF. Firmware must apply charger-policy.json.

Charge enable: default OFF
CHG_ENABLE alone cannot charge; U25 CHG_ARM is also required.

Battery-mounted thermistor
Do not substitute a fixed resistor.

Protected 103665 pack / 3000 mAh
F2 does not enforce the pack operating-current limit.

Q5 AND Q8 must both conduct to enable charging.
U25 PA0 reset/high-Z -> R79 pulls Q8 OFF.
Preload PA0 low; arm only after verified charger policy.
IWDG reset and GPIO discharge timing require bench qualification.
Supply flags identify the external protected pack through F2 and its return.

Circuit review in progress – not released for fabrication

Sheet: /Battery, thermistor and programmable charger/
File: power.kicad_sch

Title: Battery, thermistor and programmable charger

Size: A3	Date: 2026-09-24	Rev: A – engineering
KiCad E.D.A. 10.0.6		Id: 7/25

Engineering draft / named nets are common / NC = intentional unused pin

Battery, thermistor and programmable charger

Verify actual battery polarity. J2 is a battery-mounted 10k B3435 NTC. CE defaults OFF. Firmware must apply charger-policy.json.

Charge enable: default OFF
CHG_ENABLE alone cannot charge; U25 CHG_ARM is also required.

Battery-mounted thermistor
Do not substitute a fixed resistor.

Protected 103665 pack / 3000 mAh
F2 does not enforce the pack operating-current limit.

Q5 AND Q8 must both conduct to enable charging.
U25 PA0 reset/high-Z -> R79 pulls Q8 OFF.
Preload PA0 low; arm only after verified charger policy.
IWDG reset and GPIO discharge timing require bench qualification.
Supply flags identify the external protected pack through F2 and its return.

Circuit review in progress – not released for fabrication

Sheet: /Battery, thermistor and programmable charger/
File: power.kicad_sch

Title: Battery, thermistor and programmable charger

Size: A3	Date: 2026-09-24	Rev: A – engineering
KiCad E.D.A. 10.0.6		Id: 7/25

Engineering draft / named nets are common / NC = intentional unused pin

Battery, thermistor and programmable charger

Verify actual battery polarity. J2 is a battery-mounted 10k B3435 NTC. CE defaults OFF. Firmware must apply charger-policy.json.

Charge enable: default OFF
CHG_ENABLE alone cannot charge; U25 CHG_ARM is also required.

Battery-mounted thermistor
Do not substitute a fixed resistor.

Protected 103665 pack / 3000 mAh
F2 does not enforce the pack operating-current limit.

Q5 AND Q8 must both conduct to enable charging.
U25 PA0 reset/high-Z -> R79 pulls Q8 OFF.
Preload PA0 low; arm only after verified charger policy.
IWDG reset and GPIO discharge timing require bench qualification.
Supply flags identify the external protected pack through F2 and its return.

Circuit review in progress – not released for fabrication

Sheet: /Battery, thermistor and programmable charger/
File: power.kicad_sch

Title: Battery, thermistor and programmable charger

Size: A3	Date: 2026-09-24	Rev: A – engineering
KiCad E.D.A. 10.0.6		Id: 7/25

Engineering draft / named nets are common / NC = intentional unused pin

Battery, thermistor and programmable charger

Verify actual battery polarity. J2 is a battery-mounted 10k B3435 NTC. CE defaults OFF. Firmware must apply charger-policy.json.

Charge enable: default OFF
CHG_ENABLE alone cannot charge; U25 CHG_ARM is also required.

Battery-mounted thermistor
Do not substitute a fixed resistor.

Protected 103665 pack / 3000 mAh
F2 does not enforce the pack operating-current limit.

Q5 AND Q8 must both conduct to enable charging.
U25 PA0 reset/high-Z -> R79 pulls Q8 OFF.
Preload PA0 low; arm only after verified charger policy.
IWDG reset and GPIO discharge timing require bench qualification.
Supply flags identify the external protected pack through F2 and its return.

Circuit review in progress – not released for fabrication

Sheet: /Battery, thermistor and programmable charger/
File: power.kicad_sch

Title: Battery, thermistor and programmable charger

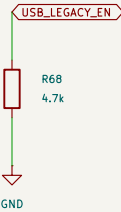
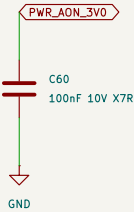
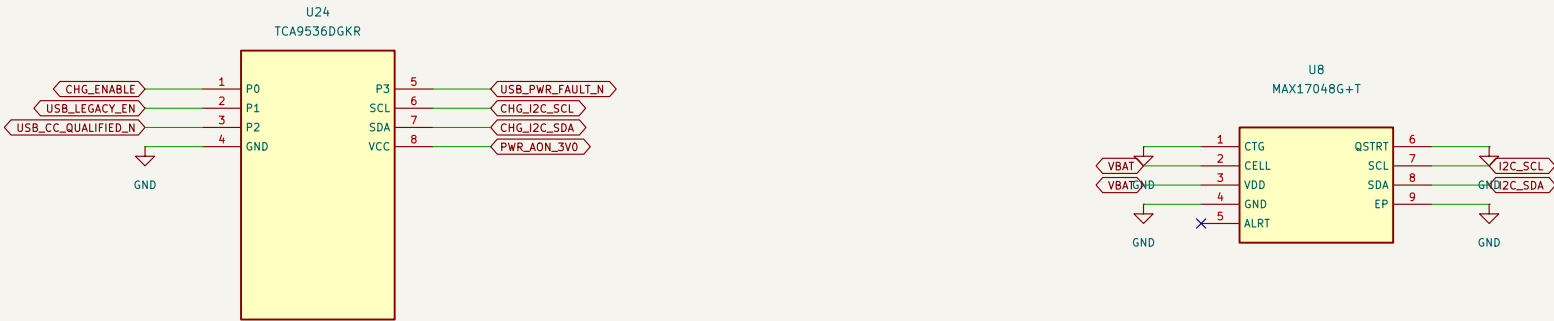
Size: A3	Date: 2026-09-24	Rev: A – engineering
KiCad E.D.A. 10.0.6		Id: 7/25

Engineering draft / named nets are common / NC = intentional unused pin

Rev: A – engineering draft
Id: 7/25

Power control and fuel gauge

I2C: TCA9536 0x41, MAX17048 0x36, charger 0x6A. Preload low output latches before enabling charger—control outputs.



Circuit review in progress – not released for fabrication

Sheet: /Power control and fuel gauge/
File: power-control.kicad_sch

Title: Power control and fuel gauge

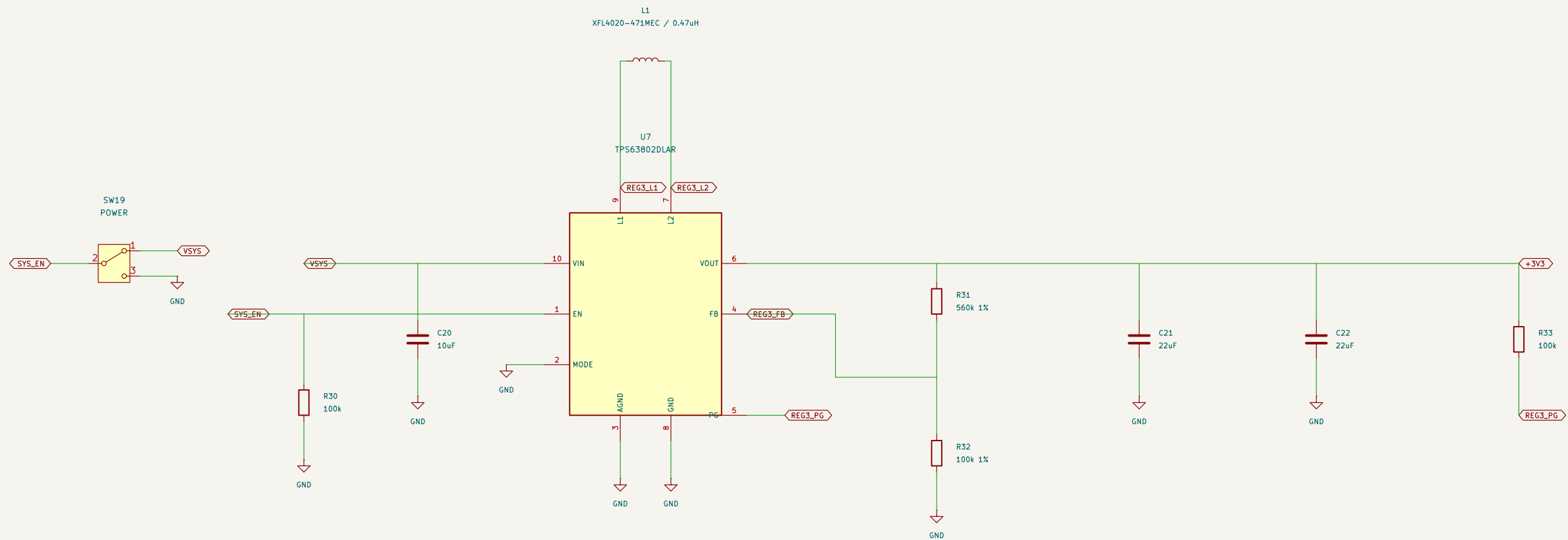
Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 8/25

Main 3.3 V buck–boost supply

TPS63802 local PCB copper is routed. External distribution, effective capacitance, thermal limits and load transients remain to qualify.



0.6 V reference $\times (1 + 560k / 100k) = 3.3$ V nominal.
Feedback senses the output capacitor node; keep it clear of the switching loop.

SW19 controls SYS_EN.
Battery / charger circuitry can remain energized while logic rails are off.

Circuit review in progress – not released for fabrication

Sheet: /Main 3.3 V buck-boost supply/
File: reg3.kicad_sch

Title: Main 3.3 V buck–boost supply

Size: A3

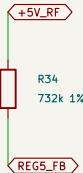
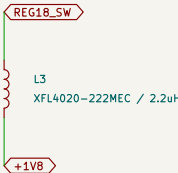
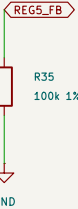
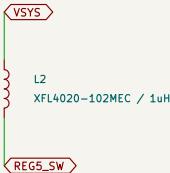
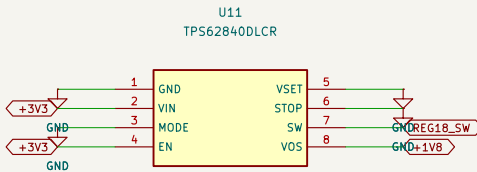
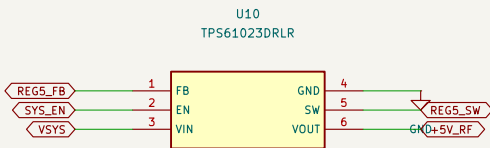
Date: 2026-09-24

Rev: A – engineering draft

Id: 9/25

Auxiliary 5 V and 1.8 V supplies

Switching loops must be short in PCB layout. The 5 V rail also supplies display backlight and the LF reader.



Circuit review in progress – not released for fabrication

Sheet: /Auxiliary 5 V and 1.8 V supplies/
File: aux-power.kicad_sch

Title: Auxiliary 5 V and 1.8 V supplies

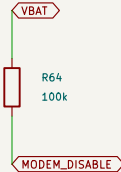
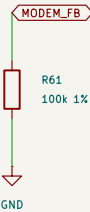
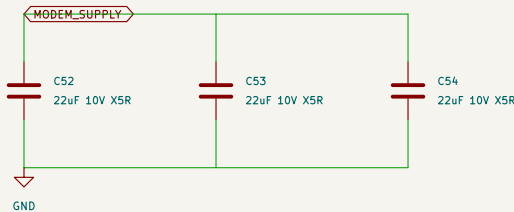
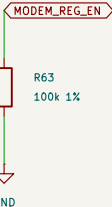
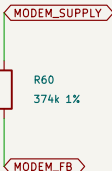
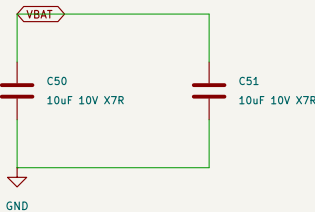
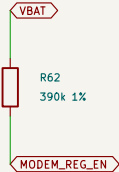
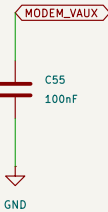
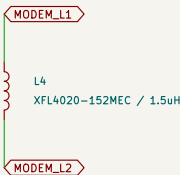
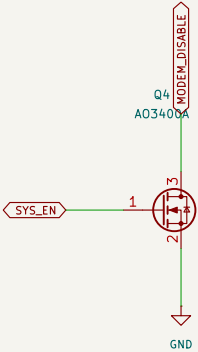
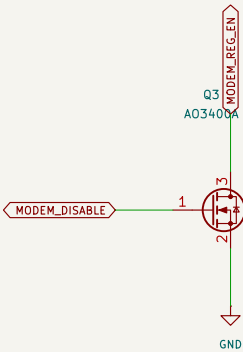
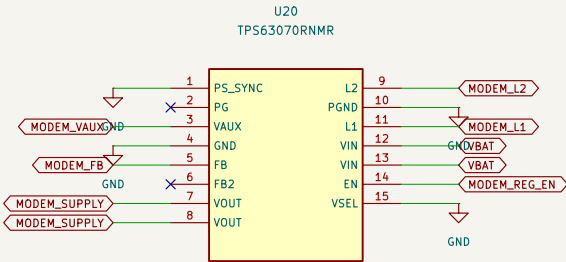
Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 10/25

Regulated modem supply and enable

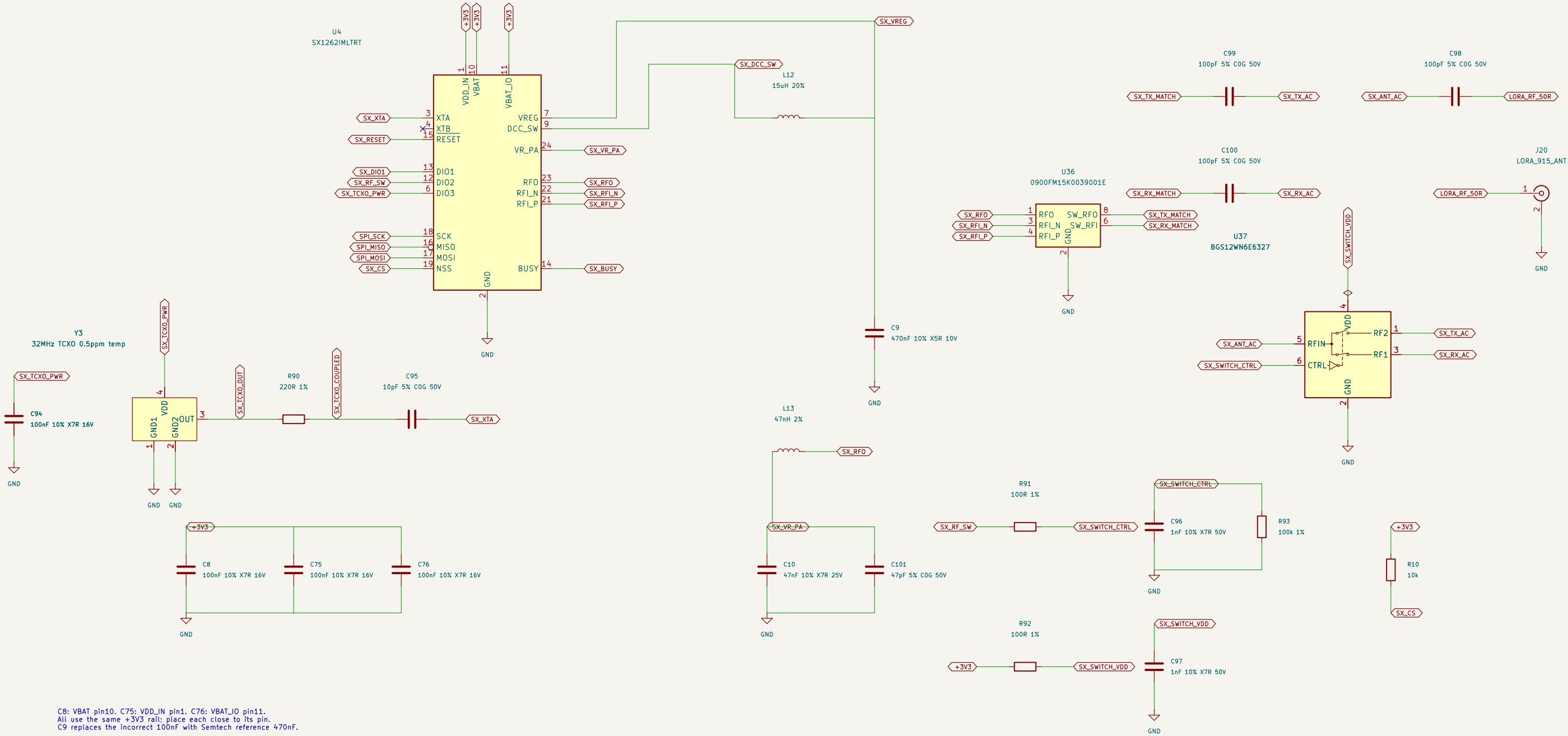
Nominal output 3.792 V. Exact modem burst current and MakerFocus pack current budget are unresolved release blockers.



Circuit review in progress – not released for fabrication		
Sheet: /Regulated modem supply and enable/ File: modem-power.kicad_sch		
Title: Regulated modem supply and enable		
Size: A3	Date: 2026-09-24	Rev: A – engineering draft
KiCad E.D.A. 10.0.6		Id: 11/25

SX1262 LoRa radio

SX1262 clock, supply, PA feed and RF matching/switch captured. RF layout incomplete; do not power.



C8: VBAT pin10. C75: VDD_IN pin1. C76: VBAT_IO pin11.
All use the same +3V3 rail; place each close to its pin.
C9 replaces the incorrect 100nF with Semtech reference 470nF.

Y3 pads 1/2 = GND, 3 = OUT, 4 = VDD. XTB is intentionally NC.
Configure DIO3 for 1.8V; provisional startup allowance 5ms.
Confirm TCXO output <=1.2Vpp; no qualified maximum in series data.

U37: CTRL=0 selects RF1/RX; CTRL=1 selects RF2/TX.
R93 defaults to RX. C98/C99/C100 explicitly block DC.
100pF RF blocks are starting values, subject to RF measurement.
No RF route or thermal/ground-via qualification in this checkpoint.

SetRegulatorMode(0x01) only in STDBY_RC, after hardware qualification.
L12 supplies the core, not the SX1262 high-power PA.
L13: 47nH PA choke; C10/C101: 47nF/47pF bypass.
RAK4270 values; final RF layout and qualification remain open.

Circuit review in progress – not released for fabrication

Sheet: /SX1262 LoRa radio/
File: radios.kicad_sch

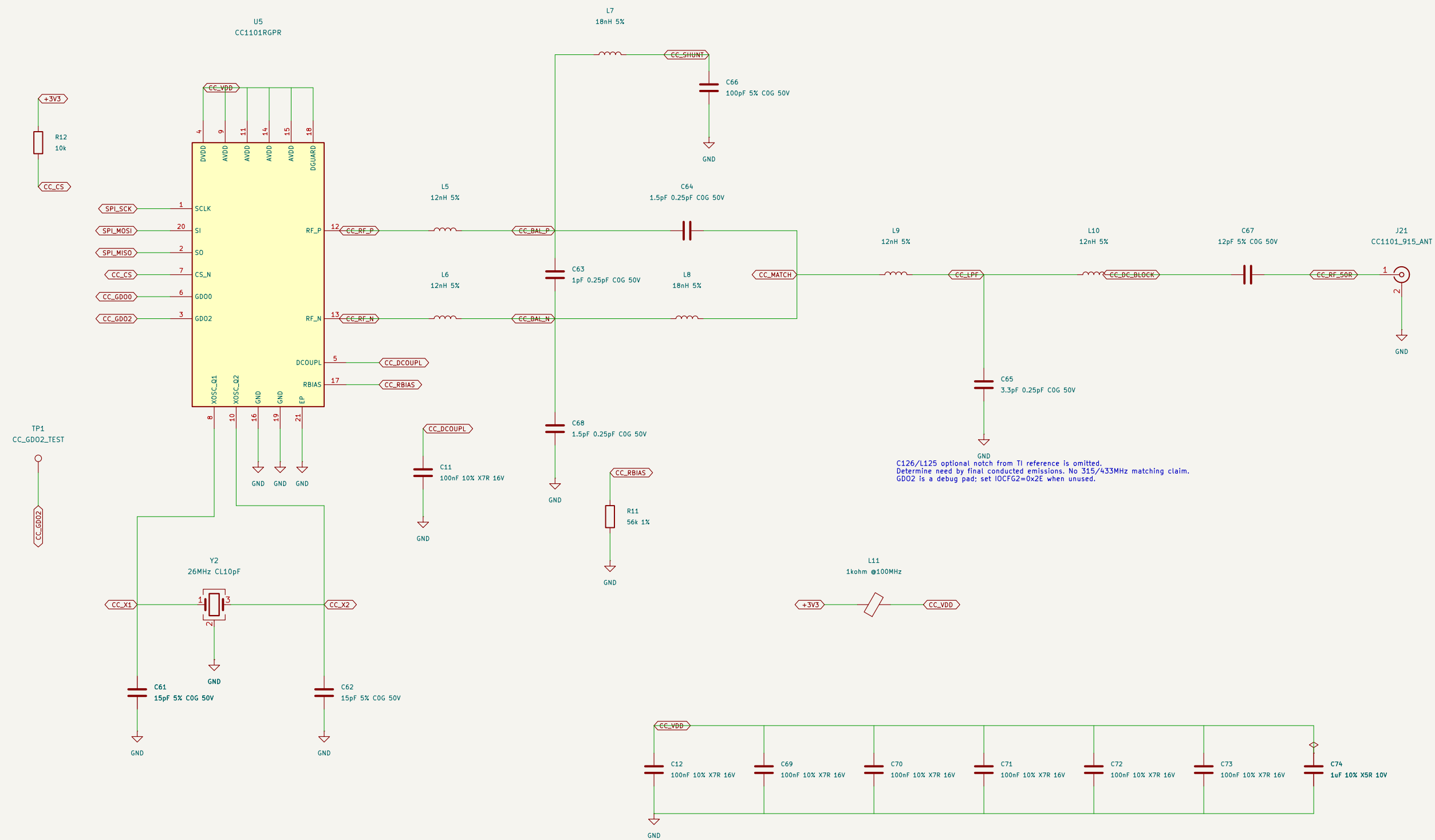
Title: **SX1262 LoRa radio**

Size: A3 Date: 2026-10-08
KiCad E.D.A. 10.0.6

Rev: **A – engineering draft**
Id: 12/25

CC1101 sub-GHz radio

868/915MHz frontend: TI SWRR045 rev3.0 / SWRS061I Fig11. RF layout and tuning are NOT qualified. Separate 50-ohm antenna.



Y2: ABM8-26.000MHZ-10-D-1-G-T
CL = 15pF / 2 + estimated 2.5pF stray = 10pF.
Measure oscillator frequency, startup and drive level.

CC_VDD flag: main +3V3 through passive L11; not an independent source.
Place one 100nF at each supply pin group, plus local 1uF bulk.

Circuit review in progress – not released for fabrication

Sheet: /CC1101 sub-GHz radio/
File: cc-radio.kicad_sch

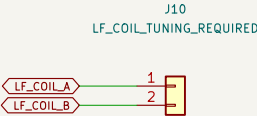
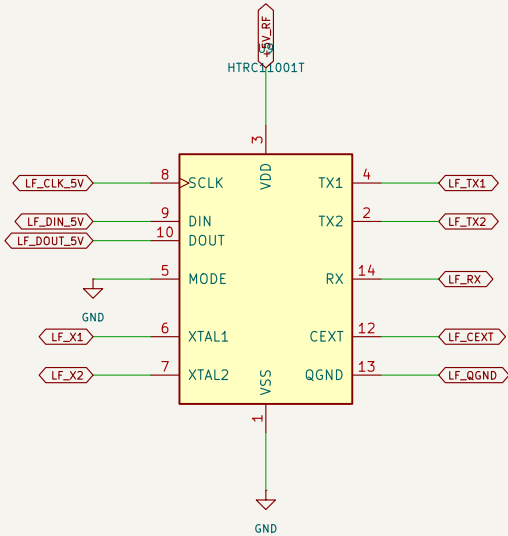
Title: CC1101 sub-GHz radio

Size: A3	Date: 2026-09-24
KiCad E.D.A. 10.0.6	

Rev: A – engineering draft
Id: 13/25

125 kHz reader analog circuit

HTRC110 requires its oscillator, QGND/CEXT bypass, resonant coil network and a timing-capable host interface.



Circuit review in progress – not released for fabrication

Sheet: /125 kHz reader analog circuit/
File: lf-radio.kicad_sch

Title: 125 kHz reader analog circuit

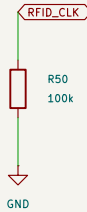
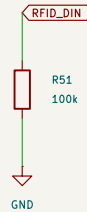
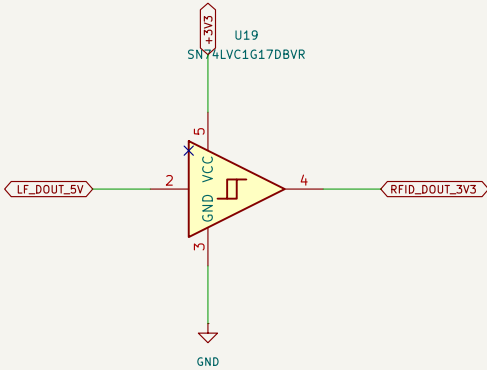
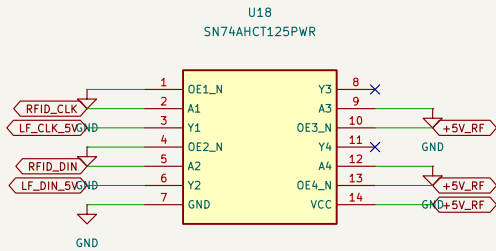
Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 14/25

LF reader logic translation

AHCT inputs accept 3.3 V logic at 5 V supply; LVC return buffer is powered at 3.3 V. Host timing remains a design requirement.



Circuit review in progress – not released for fabrication

Sheet: /LF reader logic translation/
File: lf-support.kicad_sch

Title: LF reader logic translation

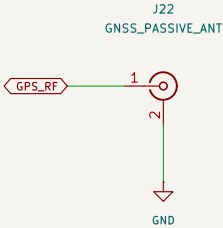
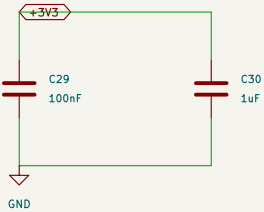
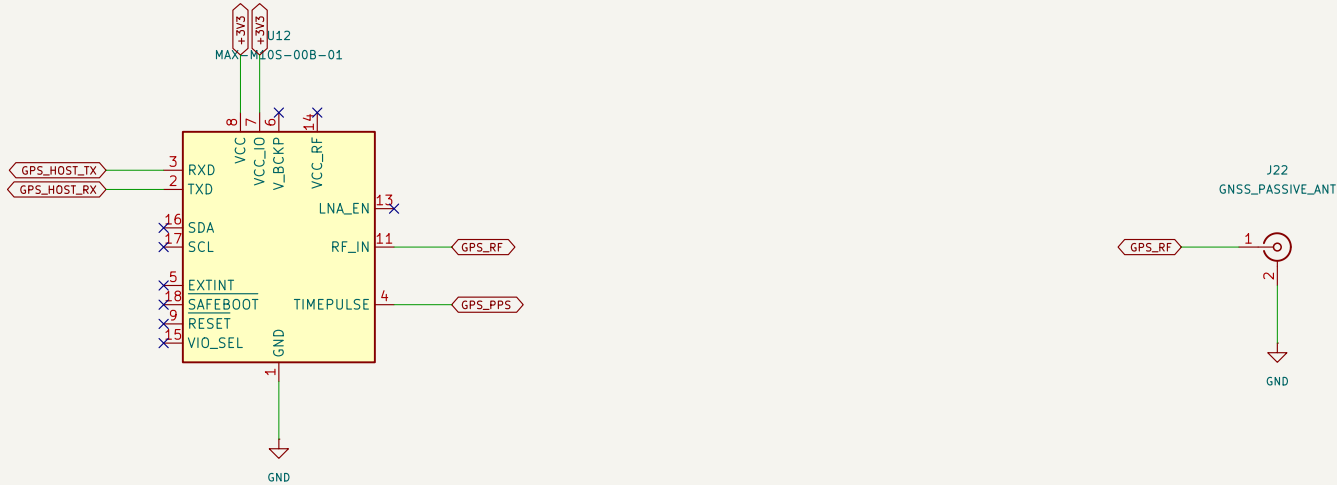
Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 15/25

GNSS receiver and passive antenna

MAX-M10S passive antenna. V_BCKP remains open: cold starts after power removal. No direct LiPo connection.



Circuit review in progress – not released for fabrication

Sheet: /GNSS receiver and passive antenna/
File: gnss-antennas.kicad_sch

Title: GNSS receiver and passive antenna

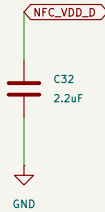
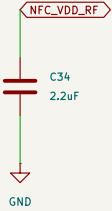
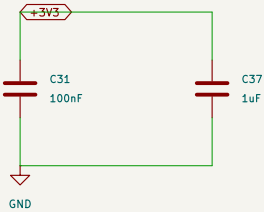
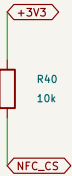
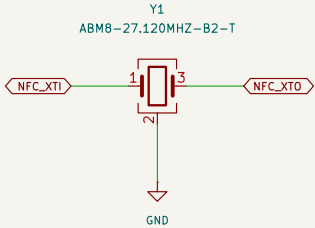
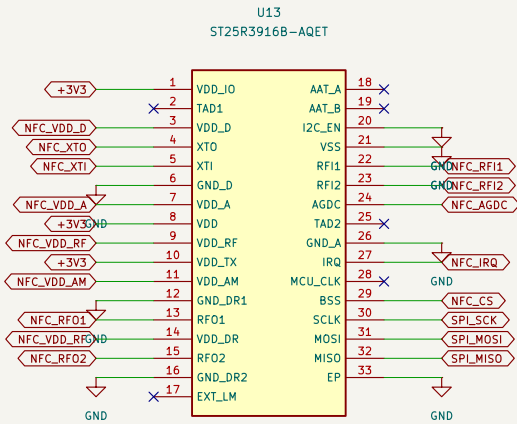
Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 16/25

NFC reader, oscillator and local supplies

ST25R3916B needs a tuned case-mounted loop and TX/RX matching network. Verify crystal load against the selected crystal.



Circuit review in progress – not released for fabrication

Sheet: /NFC reader, oscillator and local supplies/
File: nfc.kicad_sch

Title: NFC reader, oscillator and local supplies

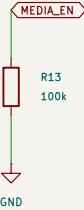
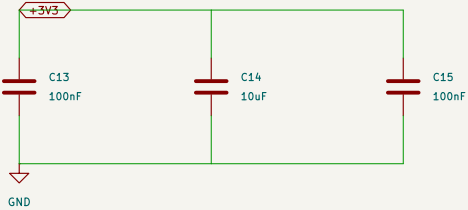
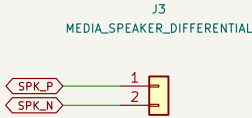
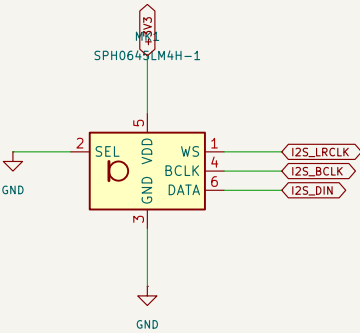
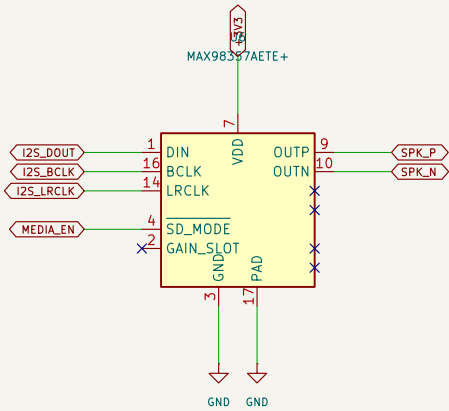
Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 17/25

Media speaker amplifier and digital microphone

Top-mounted media speaker uses differential outputs: neither speaker wire is ground. Digital microphone is case-bottom mounted.



Circuit review in progress – not released for fabrication

Sheet: /Media speaker amplifier and digital microphone/
File: audio.kicad_sch

Title: Media speaker amplifier and digital microphone

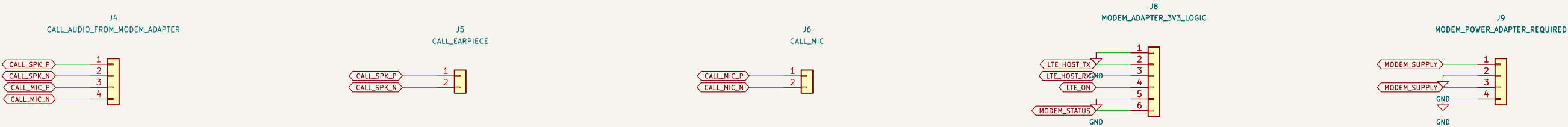
Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 18/25

Cellular adapter and call–audio harnesses

Replaceable modem adapter owns SIM, cellular RF and modem–level translation. No universal–carrier guarantee. Speaker top / mic bottom.



Circuit review in progress – not released for fabrication

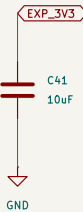
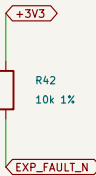
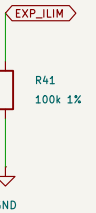
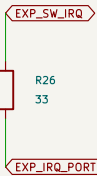
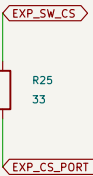
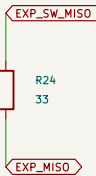
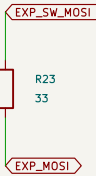
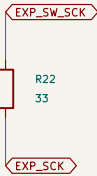
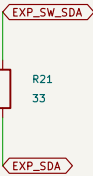
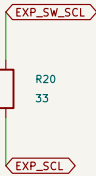
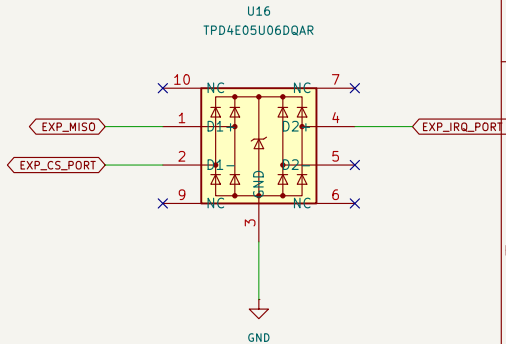
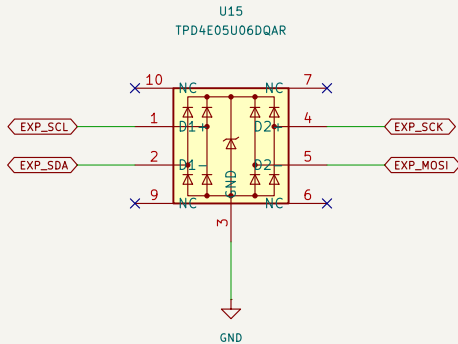
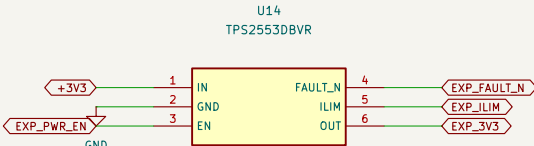
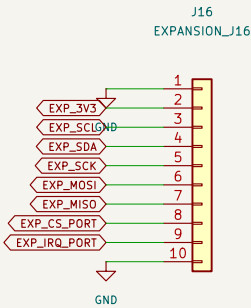
Sheet: /Cellular adapter and call–audio harnesses/
File: interfaces.kicad_sch

Title: Cellular adapter and call–audio harnesses

Size: A3	Date: 2026–09–24	Rev: A – engineering draft
KiCad E.D.A. 10.0.6		Id: 19/25

Protected external module port

Port signals pass through U34 on expansion-control. U14 EN defaults low; fault drives U33 and the hardware signal interlock.
No hot-plug approval.



Circuit review in progress – not released for fabrication

Sheet: /Protected external module port/
File: protection.kicad_sch

Title: Protected external module port

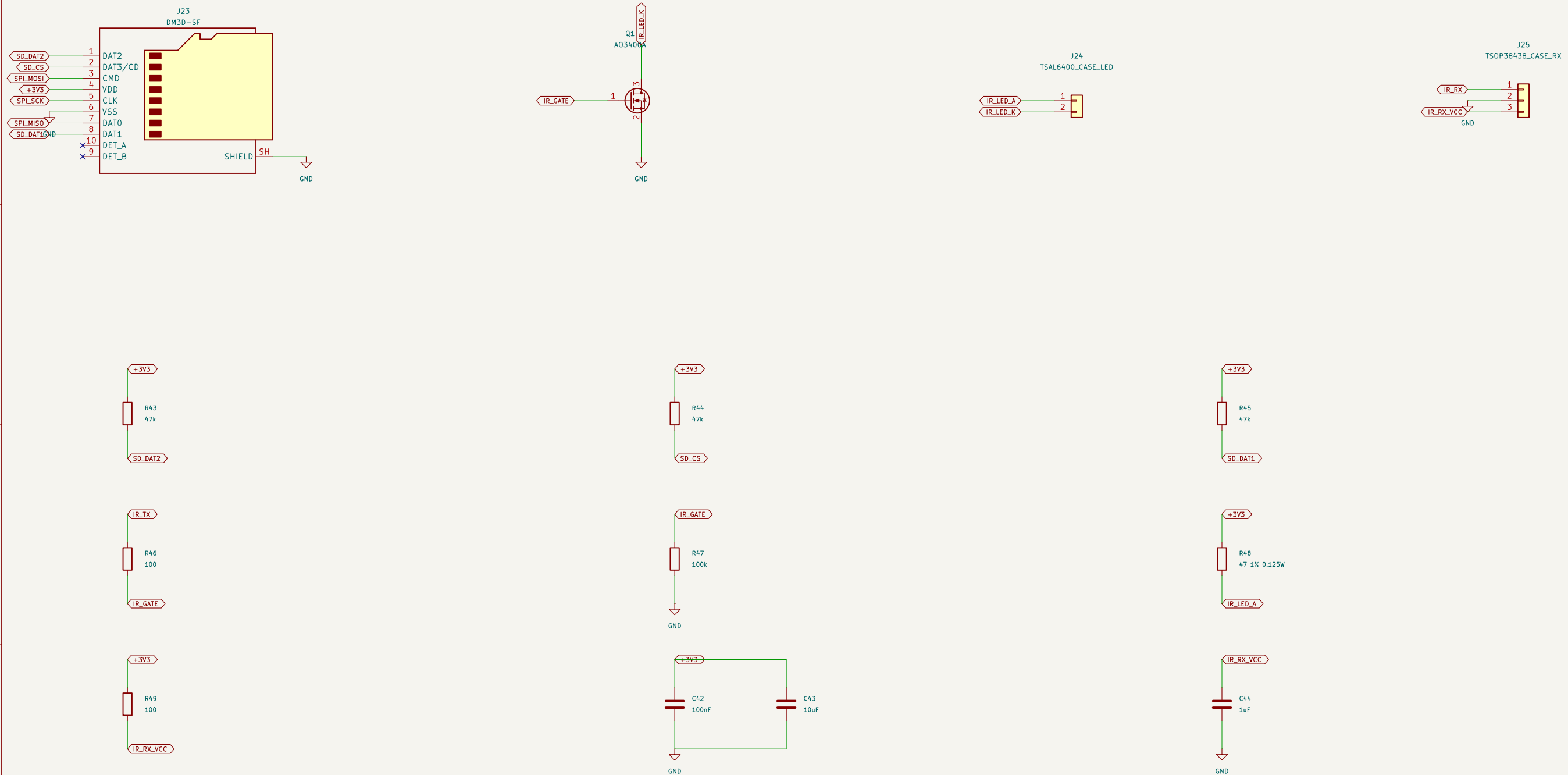
Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 20/25

MicroSD and case-mounted infrared

SPI microSD, transistor-driven IR LED, filtered receiver supply. Confirm case LED/receiver polarity and GPIO3 boot behavior.



Circuit review in progress – not released for fabrication

Sheet: /MicroSD and case-mounted infrared/
File: storage-ir.kicad_sch

Title: MicroSD and case-mounted infrared

Size: A3

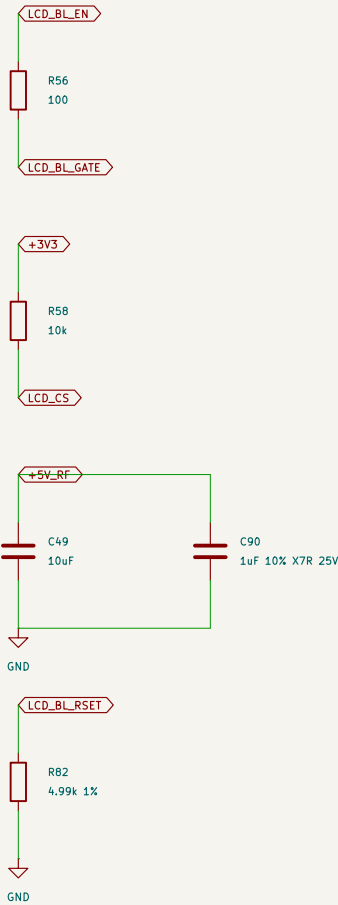
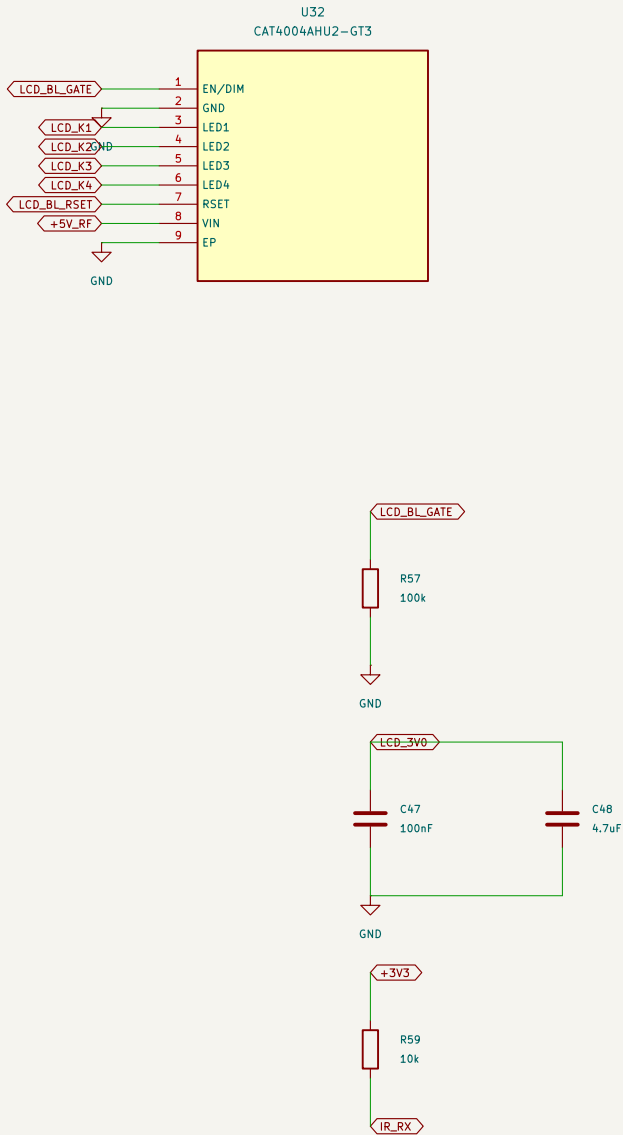
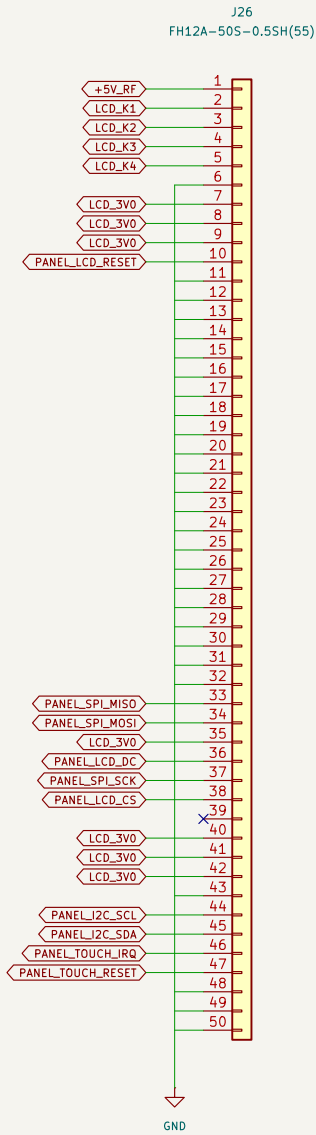
Date: 2026-09-24

Rev: A – engineering draft

Id: 21/25

External display, touch and backlight

CAT4004A: four current sinks, 4.99k sets approx. 15mA/channel. On/off only until dimming timing is qualified. Do not fabricate.



EN/DIM: external R57 pull-down retains default off.
Initialize U2 GPIO low for >=10ms before enabling.
Keep high for full-scale current; do not apply ordinary PWM.
RSET gain, current drift, heat and purchased panel remain unqualified.

Circuit review in progress – not released for fabrication

Sheet: /External display, touch and backlight/
File: display.kicad_sch

Title: External display, touch and backlight

Size: A3	Date: 2026-09-24	Rev: A – engineering draft
KiCad E.D.A. 10.0.6		Id: 22/25

Always-on charging supervisor

Separate supervisor selected. U3 remains the charger. Main host powers off at SW19. Circuit capture only; routing, STM32 firmware and fault qualification pending.

A

B

C

D

E

F

A

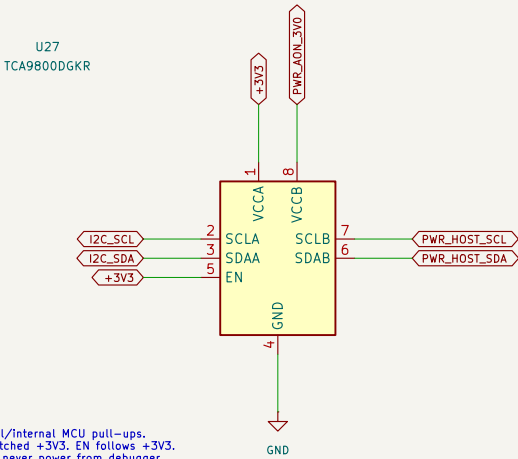
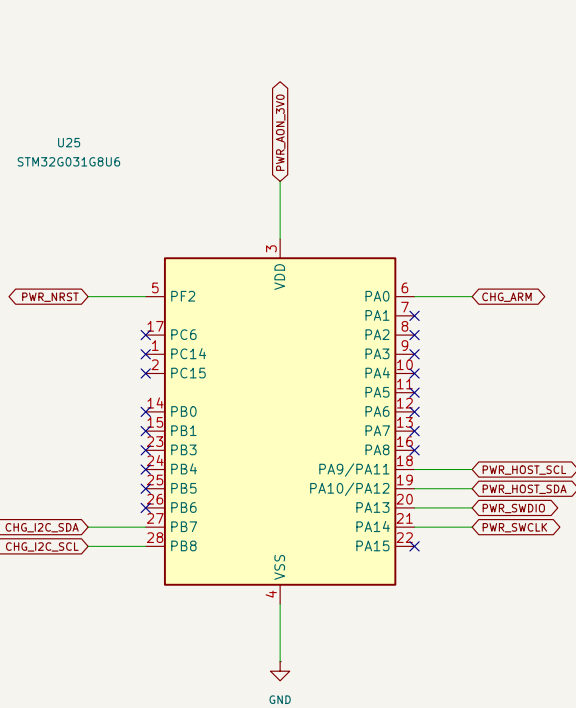
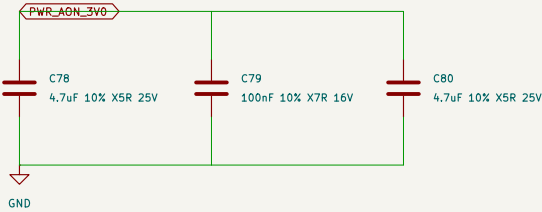
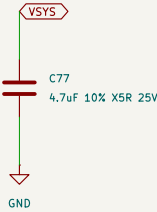
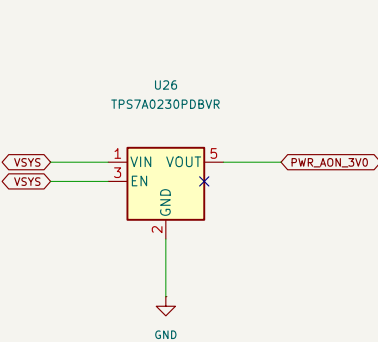
B

C

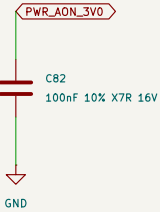
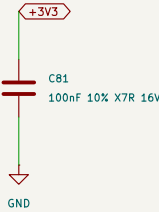
D

E

F

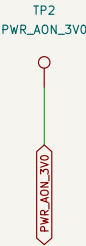
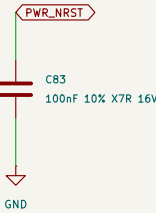
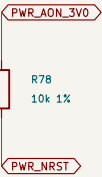
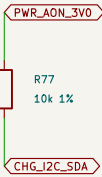
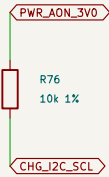


TCA9800 B side: NO external/Internal MCU pull-ups.
A-side pull-ups are on switched +3V3. EN follows +3V3.
Debug VTREF is sense only; never power from debugger.



I2C1: PB8/PB7 AF6 -> charger 0x6A + expander 0x41.
I2C2: remap PA9/PA10 to PA11/PA12, AF6.
Host slave address 0x42 reserved; protocol not implemented.

PA0: dedicated active-high charge arm. Reset analog/high-Z allows R79 at Q8 to hold charging disabled independently of U24. Never retain this output through standby or run bootloader while armed.



Circuit review in progress – not released for fabrication

Sheet: /Always-on charging supervisor/
File: power-supervisor.kicad_sch

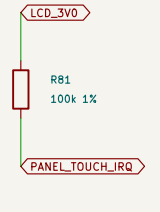
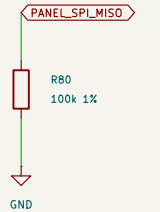
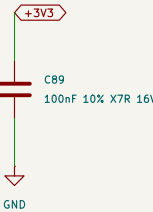
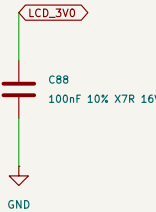
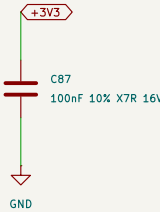
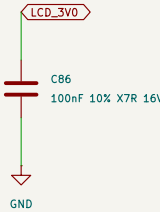
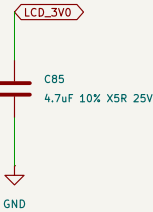
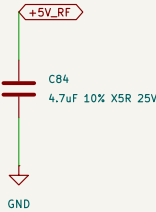
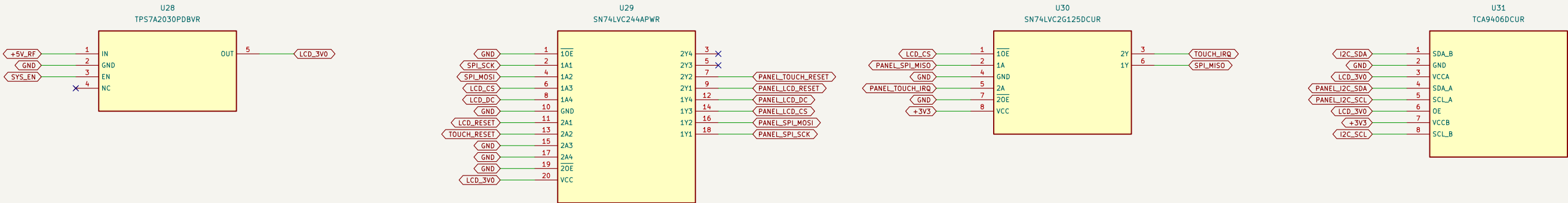
Title: Always-on charging supervisor

Size: A3 Date: 2026-09-24
KiCad E.D.A. 10.0.6

Rev: A – engineering draft
Id: 23/25

Display supply and signal translation

Interface routed and continuity checked. Qualify loads, timing, startup and case cable. Do not fabricate.



U29: 3.3 V tolerant inputs; outputs use LCD_3V0.
U30: MISO OE follows host LCD_CS; IRQ buffer always enabled.
U31: internal pull-ups; verify bus capacitance and touch option.
U28: 5 V input provides headroom; qualify load and heat.

Circuit review in progress – not released for fabrication

Sheet: /Display supply and translation/
File: display-interface.kicad_sch

Title: Display supply and signal translation

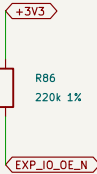
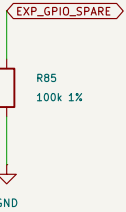
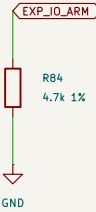
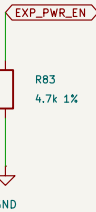
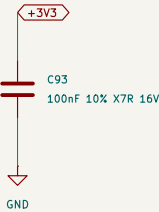
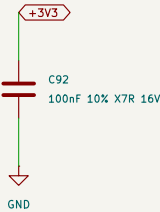
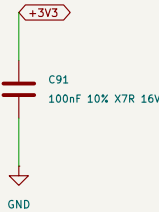
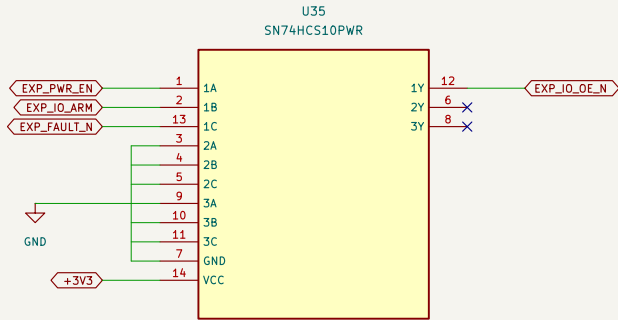
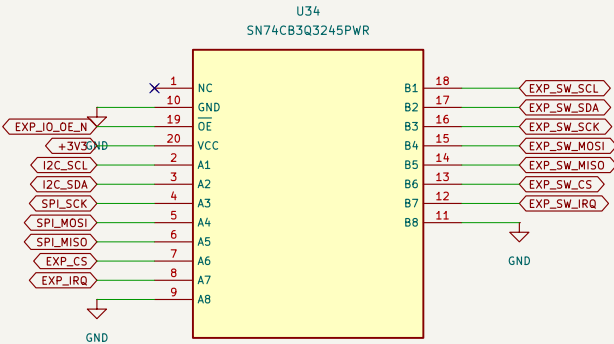
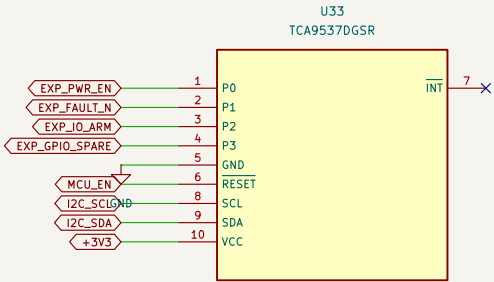
Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 24/25

Expansion power control and signal isolation

TCA9537 0x49: P0 power, P1 fault, P2 IO arm, P3 unused input. MCU_EN low resets all ports to inputs. External pull-downs retain OFF.



U35: /OE = NOT (EXP_PWR_EN AND EXP_IO_ARM AND EXP_FAULT_N).
Initialize output latch to 0xF0 before direction 0xF1. Power-only 0xF1: armed 0xF5.
Fault is polled; no MCU GPIO/strap changes. A stuck host I2C bus requires hardware reset.
No rail-good sensor; qualify load/startup and share buses only after settling with fault clear.
Accessory address 0x49 is reserved. Swap modules only with handset power off.

Circuit review in progress – not released for fabrication

Sheet: /Expansion control and isolation/
File: expansion-control.kicad_sch

Title: Expansion power control and signal isolation

Size: A3 Date: 2026-09-24
KiCad E.D.A. 10.0.6

Rev: A – engineering draft
Id: 25/25